

12.2 Partial Derivatives

$$f = f(x, y) \quad f_x(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

$$f_y(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}$$

So f_x is given by treating $y = y_0$ as a constant and differentiating by $\frac{d}{dx}$; f_y is given by letting $x = x_0$ & differentiating by $\frac{d}{dy}$.

$$\text{Notation: } f_x = \frac{\partial f}{\partial x} \quad f_y = \frac{\partial f}{\partial y}$$

$$\text{Ex } f(x, y) = x^4 y^5 \quad f_x = 4x^3 y^5 \quad f_y = 5x^4 y^4$$

$$f_x(1, 2) = 4 \cdot 1^3 \cdot 2^5 = 128 \quad f_y(1, 2) = 5 \cdot 1^4 \cdot 2^4 = 80$$

$$\text{Ex } f(x, y) = e^{xy^2} \quad f_x = y^2 e^{xy^2} \quad f_y = 2xy e^{xy^2}$$

$$f_{xx} = y^2 \cdot y^2 e^{xy^2} = y^4 e^{xy^2} \quad f_{xy} = (f_x)_y = (y^2 e^{xy^2})_y = 2ye^{xy^2} + y^2 \cdot 2xy e^{xy^2}$$

$$(f_y)_x = (2xy e^{xy^2})_x = 2ye^{xy^2} + 2xy \cdot y^2 e^{xy^2}$$

$$f_{yy} = (2xy e^{xy^2})_y = 2x e^{xy^2} + (2xy)^2 e^{xy^2}$$

$$\text{Ex } f(x, y) = \sin(x+y) \quad f_x = \cos(x+y) \quad f_y = \cos(x+y)$$

$$\text{Ex } f(x, y) = \sin(x/y) \quad f_x = \frac{1}{y} \cos(x/y) \quad f_y = -\frac{x}{y^2} \cos(x/y)$$

$$f_{xy} = \left(\frac{1}{y} \cos(x/y) \right)_y = -\frac{1}{y^2} \cos(x/y) + \frac{1}{y} \cdot \frac{x}{y^2} \sin(x/y)$$

$$\sum_x \quad f(x, y, z) = \frac{1}{(x^2 + y^2 + z^2)^{1/2}} \quad f_x = \frac{-2x}{2(x^2 + y^2 + z^2)^{3/2}} \quad f_y = \frac{-2y}{2(x^2 + y^2 + z^2)^{3/2}} \\ f_z = \frac{-2z}{2(x^2 + y^2 + z^2)^{3/2}}$$

$$\sum_x \text{ compute } (e^y \sin x)_x \quad (e^y \sin x)_{yx}$$

$$\frac{\partial}{\partial s} (\ln(s^2 - t^2)) \quad (\tan^{-1} xy)_{xy} \quad (\tan^{-1} xy)_{yx}$$

$$\frac{\partial}{\partial y} \frac{\partial}{\partial x} \left(\frac{2x - y}{xy} \right)$$

$$\sum_x \text{ compute } f_x(-1, 1) \text{ \& } f_y(-1, 1) \text{ for } f(x, y) = e^y \cosh x$$

$$\sum_x \text{ Show that } \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0 \quad \text{for } f(x, y) = x^3 y - xy^3 \\ \text{\& } f(x, y) = \ln(4x^2 + 4y^2)$$